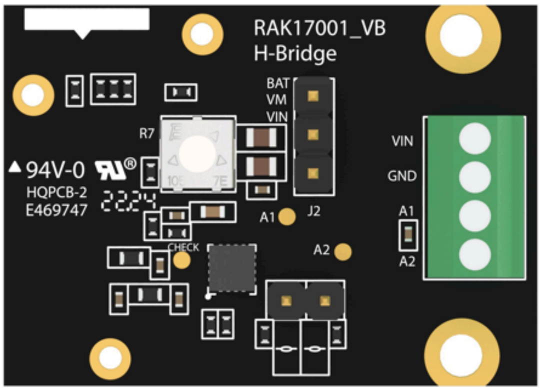
RAK17001 WisBlock H-Bridge Module Datasheet

Overview



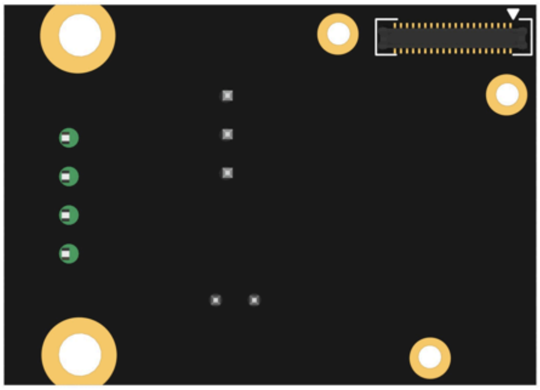


Figure 1: RAK17001 WisBlock H-Bridge Module

Description

RAK17001 is a WisBlock H-Bridge module which extends the WisBlock system with a STSPIN250 DC motor drive module from ST. It can drive dc motor and can be used as a PWM controller. Additionally, it can be mounted to IO slot of WisBlock Base board.

Features

- Sensor specifications
  - DC motor drive module
  - Operating voltage from 1.8 V to 10 V
  - Maximum output current 2.6 Arms
  - Full protection set
- Size

- 25 × 35 mm

Specifications

Overview

Mounting

Figure 2 shows the mounting mechanism of the RAK17001 module on a WisBlock Base board. The RAK17001 module can be mounted on the IO slots.

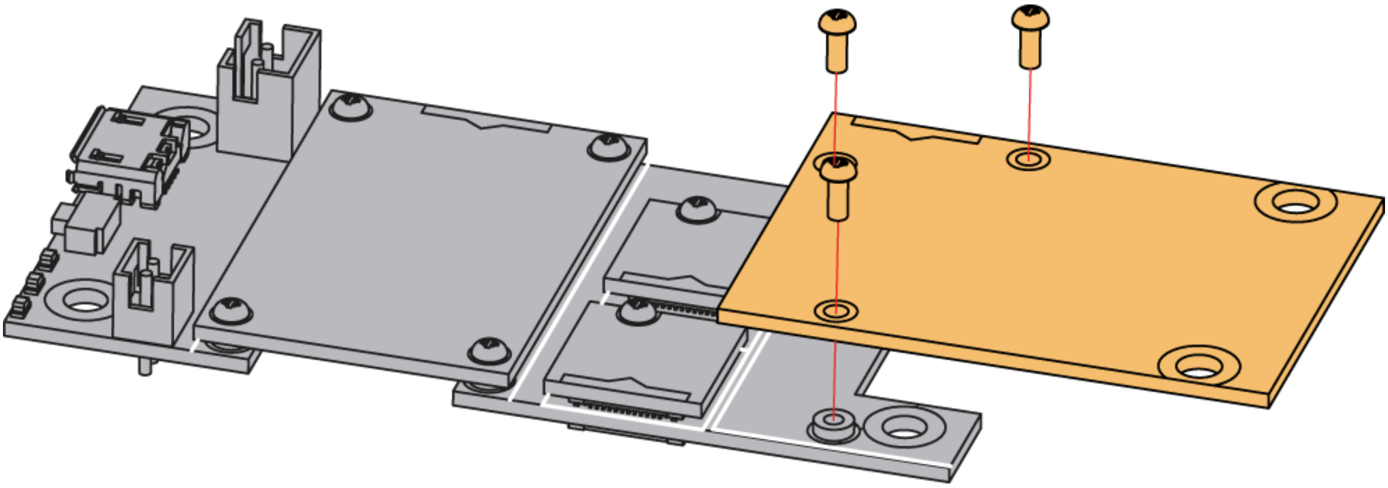


Figure 2: RAK17001 WisBlock H-Bridge Module Mounting

Hardware

The hardware specification is categorized into five parts. It shows the chipset of the module and discusses the pinouts, sensors, and the corresponding functions and diagrams. It also covers the electrical and mechanical parameters, including the tabular data of the functionalities and standard values of the RAK17001 WisBlock H-Bridge Module.

Chipset

Vendor	Part Number
ST	STSPIN250

Pin Definition

The RAK17001 WisBlock H-Bridge Module comprises a standard WisBlock connector. The WisBlock connector allows the RAK17001 module to be mounted to a WisBlock Base board. The pin order of the connector and the pinout definition are shown in **Figure 3**.

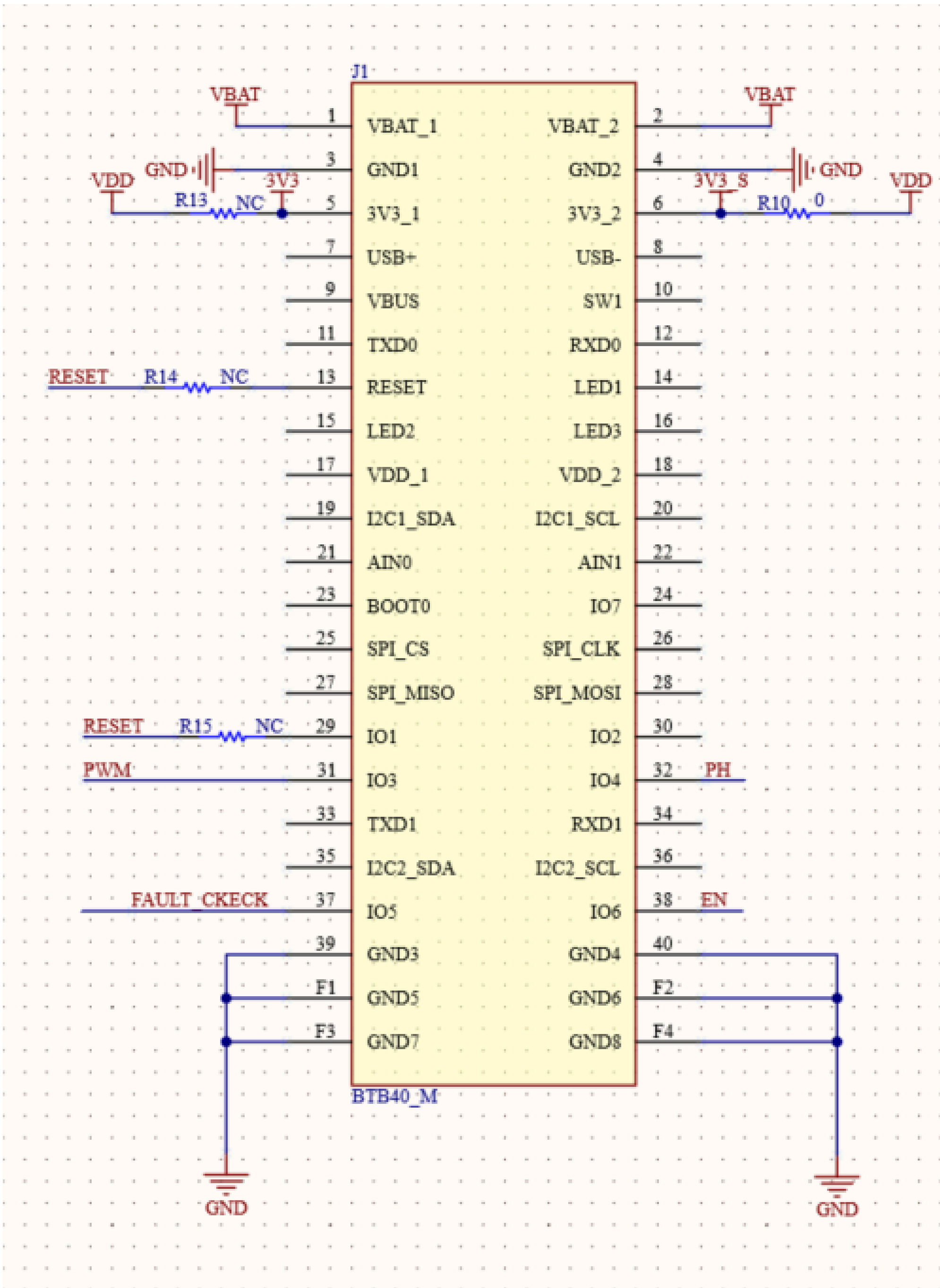


Figure 3: RAK17001 WisBlock H-Bridge Module Pinout Diagram

NOTE

- **PWM**, **PH**, and **FAULT_CKECK,EN** are connected to WisIO connector.
- **RESET** is not used by default. There is a reserved resistance for users.
- **3V3_S** is used by default for STSPIN250's logic input voltage. There is a reserved resistance for users to change to **3V3**.

Electrical Characteristics

Absolute Maximum Ratings

Parameter	Minimum	Maximum	Unit
3V3_S	- 0.3	5.5	V
VM(STSPIN250's Supply Voltage)	- 0.3	11	V

Power Supply Ratings

Symbol	Description	Condition	Min.	Nom.	Max.	Unit
3V3_S	Supply voltage	Input voltage must within this range	-	3.3	-	V
Vbat	Vbat power supply STSPIN250	normal operation	-	-	4.2	V
Vin	External power supply STSPIN250	normal operation	1.8	-	10	V

Mechanical Characteristics

Board Dimensions

Figure 4 shows the dimensions and the mechanic drawing of the RAK17001 module.

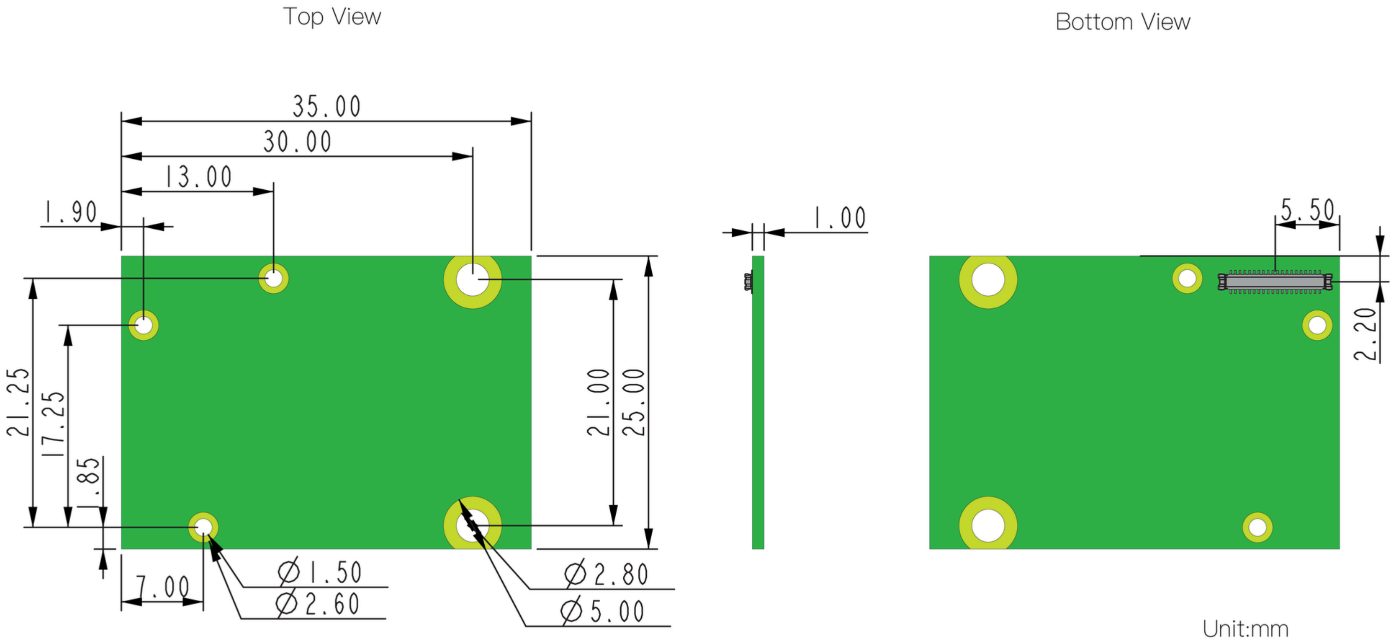


Figure 4: RAK17001 WisBlock H-Bridge Module Mechanic Drawing

WisConnector PCB Layout

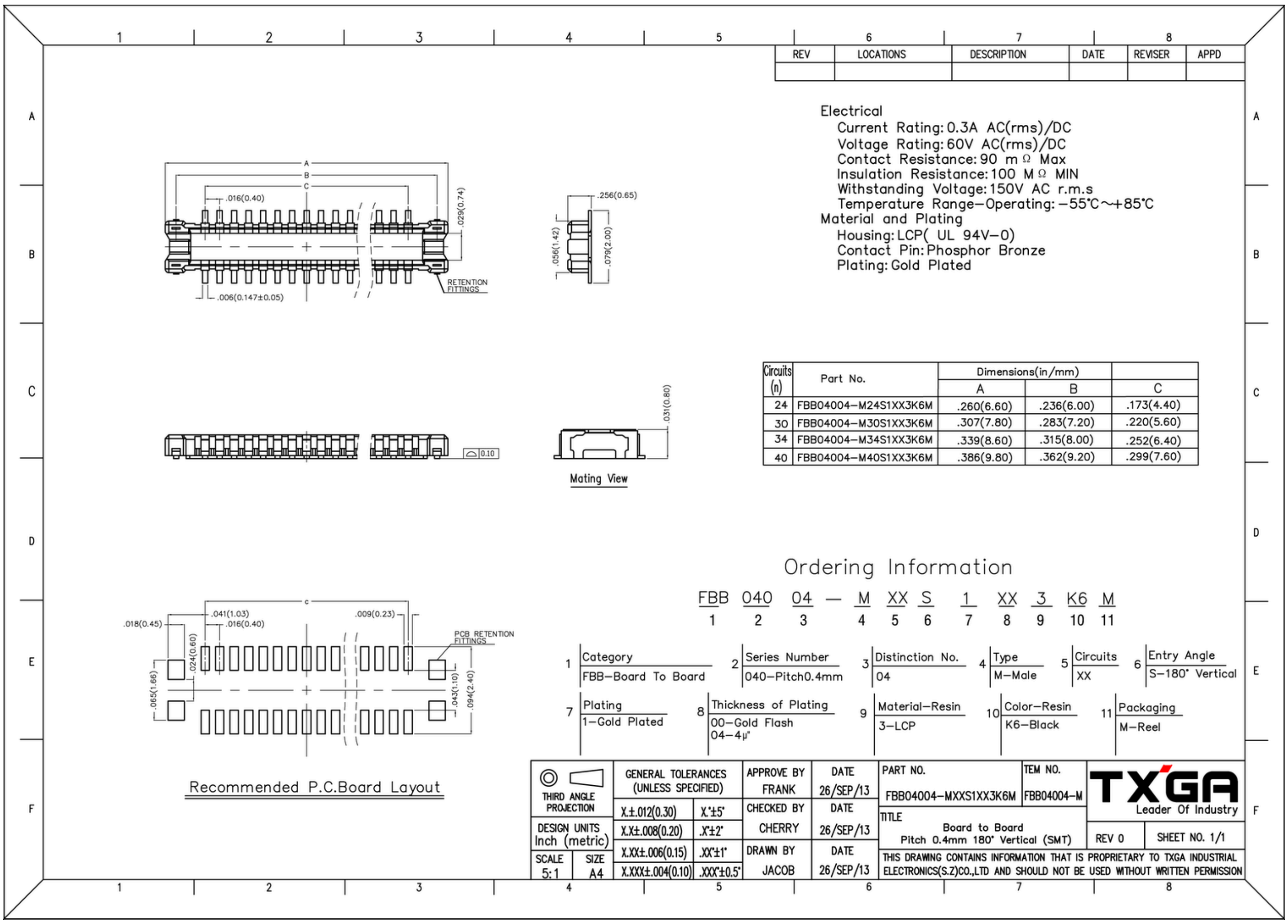


Figure 5: WisConnector PCB footprint and recommendations

Schematic Diagram

Figure 6 shows the schematic of the RAK17001 module.

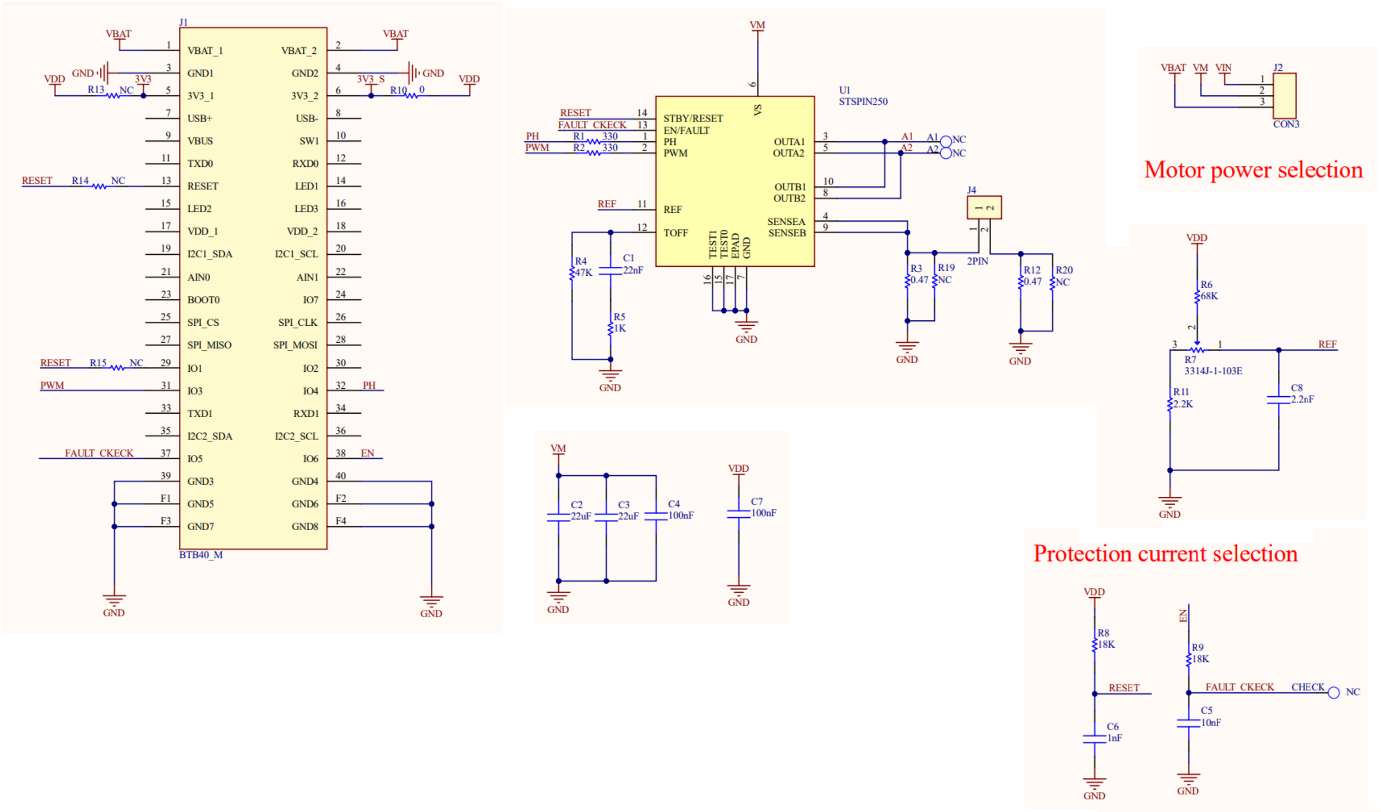


Figure 6: RAK17001 WisBlock H-Bridge Module schematics